# The Incident Is Its Own Baseline: Reference Contamination in Forensic Anomaly Calibration

**Abstract**

Unsupervised anomaly detection is routinely proposed for forensic timeline triage. Forensic use also calls for auditable operating characteristics. Both require a reference population of normal behaviour, which an examiner can rarely verify after an incident. The reference usually comes from the seized exhibit, which already contains the incident. No identifiable reference within it is known to be clean, representative and large enough. We measure the cost on Plaso timelines from five public forensic challenge exhibits, using conformal calibration over five detectors. First, context size can make calibration impossible. A leave-one-out conformal $p$-value among $n$ episodes cannot fall below $1/n$. Two of the five exhibits therefore support no claim at $\alpha = 0.001$ at any granularity. Second, controlled episode-injection experiments isolate reference composition at fixed size. On the event-log substrate, we replace each episode's same-context incident siblings with an equal number of non-injected episodes. This raises context-averaged episode-level power by 0.250 at $k = 50$ and $\alpha = 0.05$ (95% CI $[0.181, 0.330]$). Contamination also lowers the probability of discovering any part of the incident by 0.085 to 0.132. The effect persists with same-exhibit donors, with no detectable change in magnitude. Finally, deeper contexts, searches across contexts and earlier reference periods do not reliably repair the problem. Using an earlier period also sacrifices exchangeability. We release the evaluation protocol and a preflight utility that reports which contexts an exhibit can support.

*Keywords:* digital forensics, timeline analysis, anomaly detection, conformal calibration, error rates, reference contamination

## 1. Introduction

A forensic examiner facing a supertimeline with hundreds of thousands of events must decide which deserve attention [20, 17, 8]. Automated triage addresses the case backlogs motivating recent work on machine learning in digital forensics [15]. Unsupervised anomaly detection is one such approach, with substantial prior work on forensic artefacts and timelines [30, 22]. Forensic use also requires auditable operating characteristics [7]. A ranking directs attention but does not establish whether a screening threshold has a defensible interpretation [4]. This matters when the flagged episodes become an examiner's review queue.

Both ranking and calibration require a reference population of normal events [4, 12, 28]. Existing forensic timeline detectors obtain one by training on a partition assumed to be benign [30]. Outside forensics, a clean period, machine or curated corpus can usually provide this reference. Outside forensics, a clean period [27, 18], machine [23] or curated corpus [5] can usually provide this reference. Post-incident digital forensics typically offers none of these. The examiner has one seized exhibit in which normal and incident activity are interleaved, with no pre-incident image of the same machine. The reference population and suspected activity come from the same data. The incident can therefore affect what calibration treats as ordinary activity. An episode may be screened out because related incident episodes have entered its reference, not because the episode itself is less unusual.

No reference identifiable within the exhibit is known to be clean, representative and large enough [3, 1]. Using the whole exhibit preserves representativeness and size but includes the incident in the baseline. Narrowing to a forensic context increases specificity at the cost of both. The reference shrinks, and the incident's share may grow depending on how its episodes are distributed across contexts. An earlier timeline segment may precede the incident but sacrifices exchangeability because machine behaviour also drifts for unrelated reasons. [2, 14] We measure the cost of each strategy. We use five standard detectors and calibrate their scores with leave-one-out conformal $p$-values. These provide rank-based screening at a nominal level under exchangeability [33]. But calibration cannot make an undersized reference support a desired significance level [4]. Nor does a precise $p$-value show that the reference is free of the incident [3]. Both limits must be considered before the ranking is used.

Our main contributions are as follows.

- **Calibration admissibility.** We use the finite-sample resolution limit of conformal $p$-values to define admissible forensic contexts. A released preflight utility checks this criterion using only episode counts.
- **Controlled isolation of self-contamination.** A size-matched clean counterfactual separates the effects of reference size and composition.
- **Validation with same-exhibit donors and incident-level outcomes.** The effect persists after removing exhibit-identity shift and aggregating episodes to the incident level.
- **An audit of natural baseline strategies.** Matched controls distinguish each strategy's failure from sample-size or multiplicity artefacts.

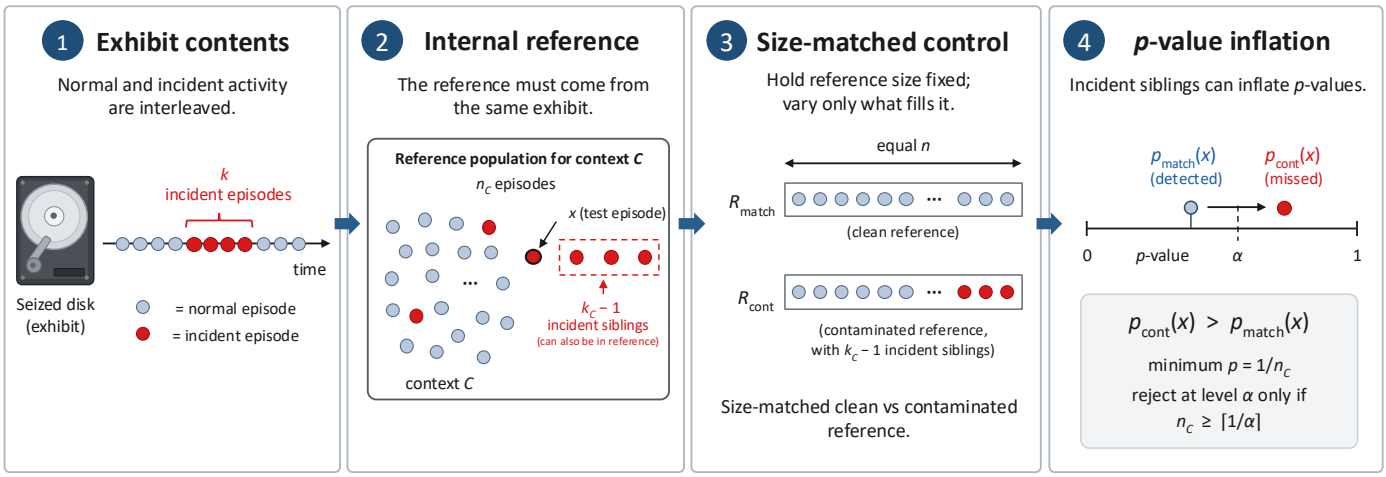



Figure 1: The argument. A seized exhibit contains both normal and incident activity (1) and supplies the reference population (2). Same-context siblings of a tested episode $x$ therefore enter its own reference. Our central experiment varies reference composition while holding size fixed (3), isolating the loss due to composition. Panel (4) illustrates masking: $p_{\text{match}}(x) \leq \alpha < p_{\text{cont}}(x)$. The episode is flagged only under the matched reference. The floor $p \geq 1/n_C$ limits the smallest $\alpha$ the exhibit can support.

A contaminated reference can reduce the power of conformal outlier detection [3]. We study the forensic setting in which the incident under investigation produces this contamination. Its severity depends on how the incident is distributed across the exhibit. The exhibit has no pre-existing ground-truth labels, so cleaning the reference requires analyst adjudication—the scarce resource triage aims to conserve.

## 2. Related work

### 2.1. Unsupervised detection on forensic artefacts

Forensic applications of unsupervised detection generally assume a usable notion of normal. Studiawan and Sohel [30] train deep autoencoders on non-malicious timeline events and use reconstruction error to flag anomalies. This requires a benign training partition. Khalid et al. [22] apply Isolation Forests, autoencoders, $k$-means, DBSCAN and Gaussian mixtures to unlabelled memory-forensic data, explaining the clusters with SHAP. They evaluate against malware-family labels unavailable in a live investigation. Both advance analysis of forensic data. Neither addresses the interpretation of anomaly scores or the error rates of decisions based on them.

### 2.2. Conformal calibration and its assumption

Conformal prediction [33] converts an arbitrary score into a $p$-value valid under exchangeability, without specifying a distribution. This makes it attractive when operating characteristics must be reported. It requires exchangeability between the test point and its reference. A forensic timeline, however, is ordered by the process under investigation. Weighted and non-exchangeable variants [32, 2] bound the coverage gap by a distributional distance. We measure temporal non-exchangeability on forensic exhibits and test whether documented incident activity explains it (Section 5.5.3).

### 2.3. Contaminated references

Masking among mutually similar outliers is a classical problem [21, 13]. Bashari et al. [3] study its conformal analogue. A small fraction of outliers in the calibration reference yields conservative type-I error control but reduces power. Their active cleaning procedure spends a limited labelling budget on the reference points most likely to be outliers. This remedy assumes a labelling oracle. We examine how contamination arises within a single post-incident exhibit, where no such oracle exists.

Conformal test martingales address the sequential problem by accumulating evidence against exchangeability as observations arrive [34]. Two concurrent works address contamination in this setting. Kuang and Xia [24] observe that a growing reference set admits post-change observations, weakening subsequent evidence of a shift. Their predictive rank martingales keep the calibration sample fixed. Bhattacharyya and Ramdas [6] develop a theory of conformal $p$-values for non-exchangeable data with an unknown changepoint. A stream can, in principle, freeze its reference before the change and use that segment as a baseline. For a seized exhibit, the entire timeline is already present and the changepoint is unknown. Its early and late portions remain non-exchangeable even after documented suspect activity is removed (Section 5.5.3). Freezing an early reference therefore does not recover validity.

### 2.4. Contextual anomaly detection

Conditioning a reference on context addresses the distinction between rarity and relevance. Contextual attributes define the reference group; behavioural attributes are scored within it. This distinction originates in conditional anomaly detection [29] and underlies the contextual category in the standard taxonomy [12]. Calikus et al. [10] show that not every candidate context is useful and select among them using active labels. Recent work combines hierarchical local and global references to avoid treating dense anomaly clusters as normal [35]. We examine whether forensic exhibits contain enough episodes to calibrate

at a useful $\alpha$ and whether contextual signal survives a matched null. These tests assess the available reference populations, not the correctness of those methods.

## 3. Method

### 3.1. Episodes

Raw timeline records are a poor unit of analysis. A supertimeline produced by log2timeline and Plaso [19] merges every timestamped artefact on the image. Machine-generated repetition dominates the result. Collapsing identical (host, channel, source, event_id) tuples within a 60 s bucket removes 81–90% of all records on our five exhibits. Without this step, a single inactivity-gap episode absorbs up to 61,770 events. Ranking records against records ranks logging behaviour rather than activity.

Aggregating low-level records into higher-level units is standard in timeline analysis [20, 8]. Our rule is deliberately simple to keep the unit reproducible without artefact-specific knowledge. An *episode* is a maximal run of collapsed records on one host separated by at most a 60 s inactivity gap. Each episode carries a behavioural feature vector (volume, duration, rate, artefact-source mix, event-type diversity and entropy, MACB mix, path diversity and depth, size statistics). The released code lists all features and detector hyper-parameters. Section 4 gives the fitting and seed policy.

### 3.2. Forensic context and the behavioural vector

A *context* groups episodes into a reference population; the behavioural vector is what is scored. Conditional anomaly detection draws this distinction [29], and the two should be kept disjoint. A context axis may be a deterministic function of features retained in the scored vector. Conditioning on that axis then removes the variation that made the episode anomalous. The resulting loss of power can be mistaken for a property of contextual calibration.

Table 1 gives the two hierarchies. They differ in how cleanly context and behaviour can be separated. On the NTFS supertimeline, every context attribute is a resource or time property absent from the scored vector. The separation is therefore exact. On the event-log substrate it is not. The dominant event family at $C_1$ and the flags at $C_3$ derive from artefact-family proportions and audit-category rates that remain in the scored vector. We therefore measure the effect of conditioning itself (Section 5.5.1) only on the NTFS hierarchy. The size-matched contrast of Section 5.2 is unaffected. Both arms share a context and differ only in which episodes populate it.

### 3.3. Calibration, and what it does and does not guarantee

Given an anomaly score $s(\cdot)$ from a detector fit outside the exhibit (Section 4), we calibrate leave-one-out within a context $C$:

$$p(x \mid C) \;=\; \frac{1 + \#\{\, j \in C,\; j \neq x : s_j \geq s(x) \,\}}{|C|}. \qquad (1)$$

Table 1: The context hierarchies, one per substrate. Each level adds one refinement. A dagger marks an attribute that is a deterministic function of features retained in the scored vector, so that conditioning on it is not independent of what is scored.

| depth | NTFS supertimeline | Windows event log |
|---|---|---|
| $C_0$ | whole exhibit | whole exhibit |
| $C_1$ | path zone | dominant event family$^\dagger$ |
| $C_2$ | allocation state | host |
| $C_3$ | time of day | privileged use, interactive user$^\dagger$ |
| $C_4$ | day type | time of day |

**Admissibility.** The smallest value Eq. (1) can take in a context of size $n = |C|$ is $1/n$. Rejection at level $\alpha$ is therefore impossible unless

$$n_C \;\geq\; \lceil 1/\alpha \rceil . \qquad (2)$$

The bound follows from the definition. Its empirical significance is that deeper contexts can become unusable as well as noisier. We call a context *admissible at* $\alpha$ when Eq. (2) holds. This describes what the calibration can support, not the admissibility of evidence in law. We report *episode coverage* as the fraction of an exhibit's episodes in an admissible context.

Within a fixed context, conformal ranks mechanically determine the in-sample rejection fraction. It therefore provides no evidence of validity.[1] We therefore read Eq. (1) as a within-exhibit screening level: a rank statement about a complete exhibit. We assess out-of-sample rejection rates only in the disjoint reference/test design of Section 5.5.3. The rank constraint binds a context as a whole, not a designated subset. The rejection fraction among injected episodes varies with the data. We report this subset quantity as power, comparing arms that share the test point, its score and its context.

## 4. Experimental setup

We use the public dataset of Windows forensic artefacts [26]. Its Plaso-derived timelines (l2tcsv) cover Magnet CTF 2019, 2020 and 2022, the NIST Data Leakage case, and DFIR Madness "Stolen Szechuan Sauce". We use both the Windows Event Log and NTFS halves of the dataset because they differ in the property under study. After collapsing and episode construction the event-log substrate yields 9,483/3,603/1,187/183/178 episodes and the NTFS supertimeline substrate 3,765/6,237/1,844/584/250. All five exhibits are analysed for admissibility and distributional structure. The three with sufficient episode counts serve as recipients in the injection experiments.

[1] When the scores are distinct, Eq. (1) assigns the $n$ members of a context $p$-values that are a permutation of $\{1/n, \ldots, 1\}$. The fraction at or below $\alpha$ is therefore $\lfloor \alpha n \rfloor / n$ whatever the data. Under the $\geq$ convention, ties can only make that fraction smaller.

Table 2: The three regimes. No single one is adequate alone; each removes a weakness of the others.

| regime | strength | weakness |
|---|---|---|
| cross-exhibit | multiplicity and composition fully controllable | donor carries exhibit-identity shift |
| same-exhibit | no exhibit-identity shift | incident still synthetic |
| answer-key aligned | documented real activity | weak labels, low power |

*Detectors.* We use Isolation Forest [25], Local Outlier Factor [9], a diagonal Gaussian mixture, PCA reconstruction and an undercomplete autoencoder. We fix standard settings before evaluation, without tuning on any target exhibit. Each detector is fit leave-target-and-donor-out, so it has never seen the exhibit it scores. It is fit once per (exhibit, donor) pair at a fixed seed. Replicates vary the injection, context and reference draws. This design isolates contamination of the calibration reference from contamination during detector fitting. Section 5.4.3 reports what happens when the detector is also fit on the target exhibit.

*Constructed ground truth.* We plant anomalies rather than annotate them, following the practice of controlled planting into real data for anomaly benchmarks [16]. The benchmark dimensions of point difficulty, relative frequency and clusteredness correspond to the detectability, multiplicity and internal similarity we vary here. We use three regimes with complementary weaknesses (Table 2). In the cross-exhibit regime, we transplant real episodes from a donor exhibit. These are distinct episodes, not copies of one point, so observed masking is not trivial swamping by a repeated value. In the same-exhibit regime, donors come from the recipient exhibit itself, removing exhibit-identity shift. In the answer-key-aligned regime, we use the published CFReDS answer key for the NIST exhibit. We automatically parsed 74 timestamped suspect actions from it.

*A property of the substrate.* A random forest trained on the behavioural feature vectors predicts an episode's source exhibit with 0.778 five-fold accuracy, against a majority-class rate of 0.492. The features therefore partly encode exhibit identity. The same-exhibit regime removes this weakness.

## 5. Results

### *5.1. What a forensic context can support*

Figure 2 shows the episode coverage of admissible contexts for each exhibit, significance level and depth. Attainable depth falls sharply as $\alpha$ decreases. Averaged over the three larger exhibits, coverage at the deepest level falls from 0.97 at $\alpha = 0.05$ to 0.36 at $\alpha = 0.001$. Forensic reporting motivates small $\alpha$: at $\alpha = 0.05$, an examiner faces some 470 of the largest exhibit's 9,483 episodes. Yet deep contexts become unavailable at the levels that practice would favour.

This constraint is strongest on small exhibits, even before they are divided into contexts. At $\alpha = 0.001$, a reference needs 1,000 episodes. The two smallest exhibits (183 and 178 episodes) cannot meet this requirement at any granularity. At $\alpha = 0.01$, neither supports more than the first level.

The substrate matters as much as the detector. On the event-log substrate, the natural actor axis is unusable: `user_sid` is absent from more than 90% of Windows event-log records. Only 18 of 9,483 episodes on one exhibit carry an interactive user. Conditioning on user type therefore reduces its reference from 9,483 to 18 in one step. The NTFS supertimeline supports the hierarchy. At $\alpha = 0.05$, coverage stays at 0.79 or above at every depth on all five exhibits and above 0.98 on the three larger ones. At $\alpha = 0.01$, the three larger exhibits stay above 0.84 at every depth. The admissibility grid thus determines whether contextual analysis is possible using only episode counts.

### *5.2. The incident degrades its own reference*

Throughout, a *clean* episode carries no injection in the current experiment; it is not independently verified benign activity. The absence of such verification on a post-incident image is the premise of this paper. We call the effect measured here *self-contamination*, the endogenous case described in Section 1. The incident dilutes its own baseline. The reference degrades in proportion to how much of the incident it absorbs. Write $\mathcal{I}$ for the injected set of an incident, $k = |\mathcal{I}|$ for its total multiplicity and $k_C = |\mathcal{I} \cap C|$ for the number of its episodes that fall in a context $C$. An incident spread over several contexts has $k_C < k$ in each. For an injected episode $x \in \mathcal{I} \cap C$, partition the clean episodes of $C$ into a base $B_C$ and a reservoir. We compare three references; the last two have identical cardinality:

$$R_{\text{base}} = B_C, \tag{3}$$

$$R_{\text{match}} = B_C \cup U_C, \qquad |U_C| = k_C - 1, \tag{4}$$

$$R_{\text{cont}} = B_C \cup ((\mathcal{I} \cap C) \setminus \{x\}), \tag{5}$$

where $U_C$ is drawn from the reservoir, so the size-matched replacement contains $k_C - 1$ episodes rather than $k - 1$. Contexts are formed at the deepest level $C_4$ of the event-log hierarchy. Within each, a quarter of the clean episodes (at least $k_C$) form the reservoir; the rest form $B_C$. Every cell is repeated over 20 injection seeds. The test point, its score, the detector, the context and the calibration formula are identical across arms. Because $R_{\text{match}}$ and $R_{\text{cont}}$ agree in size, their contrast isolates the loss due to reference composition. At $k = 1$, all three coincide, as the power curves confirm at 0.5657.

Table 3 shows that adding $k_C - 1$ clean episodes has essentially no effect at any $k$: $R_{\text{match}}$ stays within 0.01 of $R_{\text{base}}$ throughout. Adding the same number of incident episodes costs 0.250 of power at $k = 50$, pooled over heterogeneous and homogeneous injection sets. The loss therefore reflects the composition of the added episodes.

The effect extends to the screening levels motivated by forensic reporting. At $k = 50$, repeating the design at $\alpha = 0.01$, 0.05 and 0.10 gives size-matched losses of 0.187 $[0.120, 0.264]$, 0.217 $[0.153, 0.294]$ and 0.147 $[0.095, 0.216]$, respectively. Every interval excludes zero. This sweep and the detector-fitting

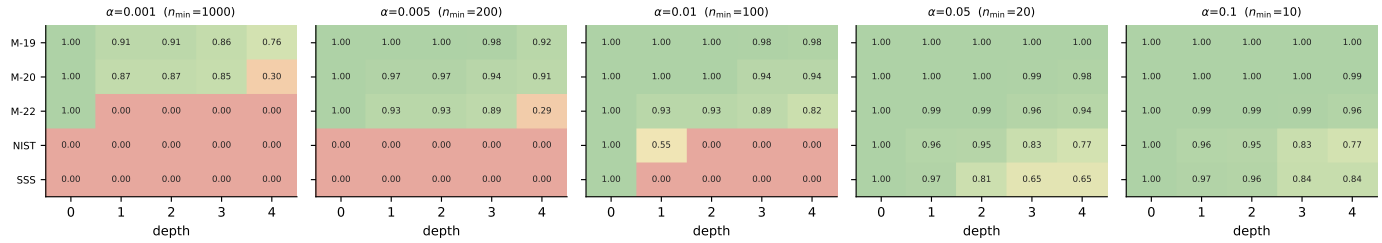



Figure 2: Admissibility grid (event-log substrate), showing the fraction of an exhibit's episodes in a context large enough to permit rejection at that $\alpha$. A red cell means no episode at that depth can be flagged, however anomalous.

Table 3: Self-contamination, isolated by a size-matched control. Cross-exhibit regime, Windows event-log substrate, episode-level power at $\alpha = 0.05$. Clustered bootstrap over (exhibit, donor, detector), 3000 resamples, pooled over heterogeneous and homogeneous injection sets. The unit of averaging is the (injection set, context) pair, so a context holding one injected episode counts as much as a context holding many. The full multiplicity sweep is in Figure 3(a).

| $k$ | $R_{\text{base}}$ | $R_{\text{match}}$ | $R_{\text{cont}}$ | matched loss (95% CI) |
|---|---|---|---|---|
| 1 | 0.566 | 0.566 | 0.566 | 0.000 |
| 5 | 0.594 | 0.594 | 0.524 | 0.069 [0.021, 0.133] |
| 20 | 0.666 | 0.660 | 0.502 | 0.157 [0.099, 0.234] |
| 50 | 0.676 | 0.681 | 0.431 | 0.250 [0.181, 0.330] |

Table 4: Incident-level discovery, size-matched. Cross-exhibit regime, Windows event-log substrate, $\alpha = 0.05$. Here, hit is the probability that at least one episode of the incident is flagged; flag is the number flagged. Subscripts $m$ and $c$ denote the size-matched clean and the contaminated reference. Both are averaged over trials, each trial pooling the episodes of one incident across the contexts they fall in. Injected episodes landing in a context too small to calibrate are not evaluated. Thus, flag is not $k$ times the power in Table 3, which averages over contexts instead.

| $k$ | $\text{hit}_m$ | $\text{hit}_c$ | $\text{flag}_m$ | $\text{flag}_c$ | loss (95% CI) |
|---|---|---|---|---|---|
| 10 | 0.828 | 0.734 | 5.2 | 3.7 | 0.095 [0.024, 0.178] |
| 20 | 0.915 | 0.782 | 10.9 | 6.3 | 0.132 [0.058, 0.220] |
| 50 | 0.927 | 0.843 | 27.2 | 10.6 | 0.085 [0.028, 0.151] |

comparison in Section 5.4.3 use only heterogeneous injection sets. The corresponding value from Table 3 at $\alpha = 0.05$ and $k = 50$ is therefore 0.217, rather than the pooled 0.250. The episode-granularity check in Section 7 uses only injections that share a context with at least one sibling, giving 0.294. All three analyses use the same estimand; only the injections entering the average differ.

### 5.3. The loss at the incident level

An investigation depends on incident discovery, not only episode-level power. Discovery requires at least one flagged episode, and larger incidents offer more opportunities. A fall in per-episode power need not therefore reduce discovery. Under the contaminated reference alone, discovery rises with $k$; larger incidents are not simply harder to find. The relevant comparison uses a size-matched clean counterfactual. It holds the incident, its episodes and reference cardinality fixed, varying only whether siblings enter the reference.

Contamination costs between 0.085 and 0.132 of incident-level discovery, with intervals excluding zero (Table 4). At $k = 50$, it more than halves the number of incident episodes that reach the examiner. Self-contamination therefore does more than dilute scores: it reduces the probability of finding the incident at all.

*Internal homogeneity.* Under unmatched selection, the loss appeared larger for incidents whose episodes resemble one another. That selection, however, confounds internal dispersion with baseline difficulty. Matching incident sets on anomaly score leaves a pooled contrast of +0.035 $[-0.012, 0.077]$. The difference is also absent in the same-exhibit regime. We therefore do not treat homogeneity as an independent moderator.

### 5.4. Robustness of the main effect

The three analyses below test whether the effect in Section 5.2 depends on incident construction or detector fitting. They assess robustness rather than introduce new claims.

#### 5.4.1. Same-exhibit donors

A concern with Section 5.2 is that transplanted episodes may be anomalous because they come from another machine. The measured 0.778 exhibit-identity accuracy supports this concern. We therefore repeat the design with donors from the recipient exhibit itself, removing exhibit-identity shift by construction. Each same-exhibit donor's native occurrence is excluded from the reference for that trial.

Removing cross-exhibit identity shift leaves the effect intact, with no detectable change in its size. The two multiplicity curves overlap (Figure 3c), and every paired difference has an interval containing zero. At $k = 50$, the difference is +0.013 $[-0.024, 0.054]$. This interval admits a few percentage points either way. The result is therefore a failure to detect a difference, not evidence of equivalence.[2] The effect in Section 5.2 thus does not require the cross-exhibit differences introduced by transplantation.

The magnitude differs across substrates: 0.250 at $k = 50$ on the event log against 0.578 on the supertimeline. These measurements also differ in context hierarchy and reference

[2]The regimes also differ slightly in detector fitting. In the same-exhibit regime, the detector is fit on the four exhibits other than the target. In the cross-exhibit regime, it is fit on the three other than the target and donor. The near-zero paired contrast is therefore a robustness result, not a formal equivalence test.

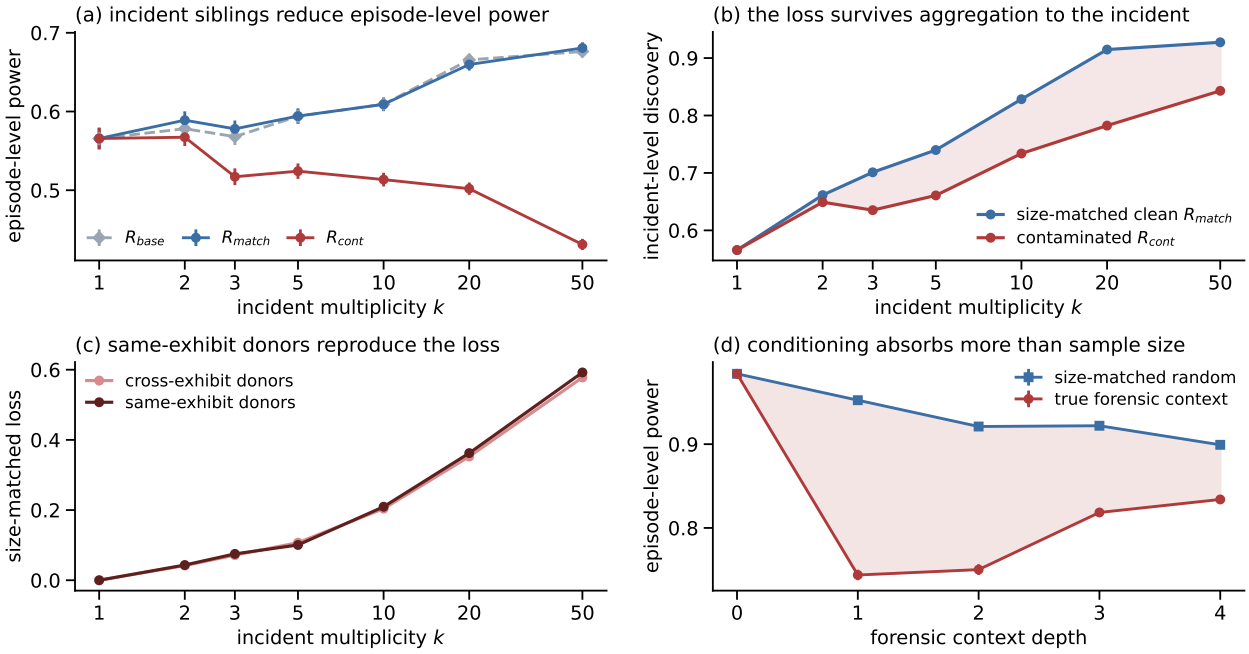



Figure 3: Panels (a) and (b) use the Windows event-log substrate, (c) and (d) the NTFS supertimeline. Error bars are standard errors. (a) Episode-level power against the three references of Section 5.2. The arms coincide at $k$=1, where no siblings exist. (b) The same two references at incident level, from the paired analysis of Section 5.3. The band is the incident-level loss. (c) Size-matched loss when donors come from another exhibit and from the recipient exhibit itself. (d) Power under the true forensic context against a reference of identical size drawn at random from the whole exhibit. The band is the part of the loss that sample size does not explain.

depth. Magnitudes are comparable only within an experiment, so each result identifies its substrate.

#### 5.4.2. *Answer-key-aligned episodes*

For the NIST Data Leakage exhibit the published answer key documents the suspect's actions with timestamps. We parsed 74 actions automatically, spanning 2015-03-23 13:29 to 2015-03-25 11:30. We define answer-key-aligned episodes as those whose interval overlaps a window around a documented action. This alignment does not identify incident episodes: it also includes concurrent ordinary operating-system activity. The primary window (±10 min, 28 aligned episodes) and the sensitivity windows (±5 and ±30 min, 19 and 63) were fixed before the analysis was run.

This analysis did not provide statistically conclusive evidence. The size-matched loss at the primary window is +0.0092 $[-0.0022, 0.0257]$. Estimates at both sensitivity windows are also positive and grow with the window, but every interval includes zero. Low baseline detectability (0.083 under a clean reference) limits the possible absolute loss. We treat this regime as an ecological sanity check on the effect's direction, not an estimate of its size.

#### 5.4.3. *Contamination of the detector as well as the reference*

Our main design excludes the exhibit from detector fitting and contaminates only the calibration reference. Fitting on the target exhibit is the more usual unsupervised arrangement and opens a second contamination pathway. Repeating Section 5.2 under this arrangement preserves the calibration effect. The size-matched loss at $k = 50$ is 0.245, against 0.218 under external fitting. The difference varies by detector in both direction and size. We therefore report a robustness check without claiming an ordering between the regimes.

Table 5: Where each reference strategy fails. Costs are measured in the section named and are not comparable across rows, which differ in substrate and design.

| reference | what goes wrong |
|---|---|
| whole exhibit | the incident enters its own baseline (§5.2) |
| deepest admissible context | a smaller reference can raise the incident's share of it, depending on how the incident spreads across contexts (§5.5.1) |
| search across contexts | no excess detection against a matched null, but an uncorrected minimum-$p$ rule raises the rejection rate (§5.5.2) |
| earlier block | the early period is not exchangeable with the late one (§5.5.3) |

### 5.5. *Natural remedies for the reference*

We test three alternative reference strategies from Section 1. Each uses a control to distinguish failure from a sample-size or multiplicity artefact. Table 5 summarises the results.

#### 5.5.1. *Localising the reference*

Given Eq. (2), a natural strategy is to calibrate in the deepest admissible context. For a fixed number of incident siblings, a smaller reference raises their share of the calibration population. Whether deeper conditioning worsens contamination therefore depends on the incident's distribution across contexts.

The penalty is graded. At total incident multiplicity $k = 20$, fixed-depth calibration loses 3% of its power at $C_0$, 21% at $C_1$–$C_2$, 32% at $C_3$ and 43% at $C_4$ on the event-log hierarchy. The deepest-admissible rule loses 42% there and 16% on the NTFS supertimeline. It therefore incurs substantial masking in these experiments. The extent depends on how many incident siblings remain in the selected context.

Localising also reduces signal independently of contamination. To isolate this effect, we calibrate each injected episode at depth $d$ against two references of size $n_d$: its true context and episodes drawn uniformly from the root (Figure 3d). This holds size fixed and varies only conditioning. The arms agree exactly at $C_0$, where their references are identical. They diverge at every deeper level, by as much as 0.21 at $C_1$. Conditioning therefore absorbs signal beyond the effect of reduced sample size, mostly at the first conditioning step.

*5.5.2. Searching across contexts*

On the NTFS supertimeline, we test the premise that some anomalies are unremarkable globally but extreme within the right context. Transplanted episodes are largely global anomalies and test this premise poorly. We therefore construct contextual anomalies by definition. Each non-injected episode is calibrated twice with its score unchanged: against its own context and against a different admissible context at the same depth. Behaviour from one context's distribution is thus judged against another's. At $C_0$, relocation changes nothing: both arms use the same reference and their rates agree at 0.0531.

The lift grows with depth, reaching 1.38× the own-context rate at $C_4$. However, the pooled excess detection rate is 0.0128, with a 95% interval of $[-0.0012, 0.0285]$ that includes zero. Per-detector estimates were heterogeneous, with no consistent direction across evaluation constructions. The two detectors that gain under relocation (Figure 4c) show no gain under a minimum-$p$ search relative to the same statistic on held-out clean episodes. We do not interpret individual detector intervals as significance statements, since five are examined. Without a matched null, the minimum-$p$ rule raises rejection on held-out clean episodes to 0.186, against a nominal 0.05.

*5.5.3. Moving the reference backwards in time*

An earlier block of the NTFS supertimeline is a candidate reference for a later one. Using only non-injected episodes, we vary the split while holding the data, detector, contexts and calibration fixed (Figure 4(a)). Reference and test episodes are disjoint, so these are out-of-sample rejection rates. We do not label them false-positive rates. Non-injected episodes are clean only in the experimental sense of Section 5.2. The exhibits contain real activity, so a rejection need not be an error. The comparison shows that the split alone changes the rate. It concerns exchangeability rather than correctness.

A block split across the whole span yields 2.66× the nominal rejection rate, against 0.90× for a random split. Even within the early half of the timeline, the block split raises the rate to 1.43× nominal. Temporal non-exchangeability is therefore already present within that restricted span. The late half and whole span raise the rate further. Half-timeline splits use smaller references and coarser $p$-values, so their rates are not directly comparable with the whole-span result. The drift is large. We measure the energy distance between behavioural feature distributions in the first and last thirds of an exhibit [31]. It exceeds the distance between two random halves of that exhibit by one to two orders of magnitude (Figure 4b). The most distant pair of thirds varies by exhibit, consistent with lifecycle phases rather than a single incident-driven change. Inflation also varies by detector, from 0.053 for LOF to 0.229 for the Gaussian mixture.

If incident activity in the non-injected set caused the inflation, removing it should reduce the rate. The NIST exhibit is the only case with a published answer key. We removed the 28 episodes overlapping a documented suspect action and repeated the comparison. Inflation persists: the block split rises from 3.3 to 3.9 times nominal (+0.026 [0.013, 0.037] over detectors and depths), while the random split stays at or below nominal in both arms. Removing episodes aligned with documented suspect actions therefore does not explain away the exchangeability failure on this exhibit.

## 6. Discussion

These results concern the evaluation of forensic anomaly detection rather than any attack class. Before reporting detector performance, studies should identify the reference population supporting their claims and verify its usability. Five practices follow from our measurements.

- **Report admissibility.** Before any $\alpha$-level claim about a context, report $n_C$ and check $n_C \geq \lceil 1/\alpha \rceil$. Report the $(\alpha, \text{depth})$ grid rather than one operating point.
- **Stress-test reference contamination.** Sweep incident multiplicity. The problem does not arise for singleton anomalies.
- **Evaluate at the incident level.** Episode-level power and incident-level discovery change differently. Only incident-level discovery answers the investigative question.
- **Do not assume that more local means a better baseline.** The two costs of a deeper context are smaller samples and local contamination. Separating them requires a size-matched control.
- **Do not assume that earlier means clean.** In our five exhibits, an earlier block was not an exchangeable baseline. A random split stayed close to nominal.

*Operational interpretation.* The quantities above translate into review effort. Screening the largest event-log exhibit at $\alpha = 0.05$ can place some 470 of its 9,483 episodes before an examiner. That is the size of the queue, not a measurement of what it contains. The separate injection experiments compare contaminated references with size-matched non-injected references. Contamination lowers the probability of surfacing any part of the incident by 9 to 13 percentage points. It also removes more than half of the injected episodes that would otherwise have been flagged. These episodes never reach the review queue.

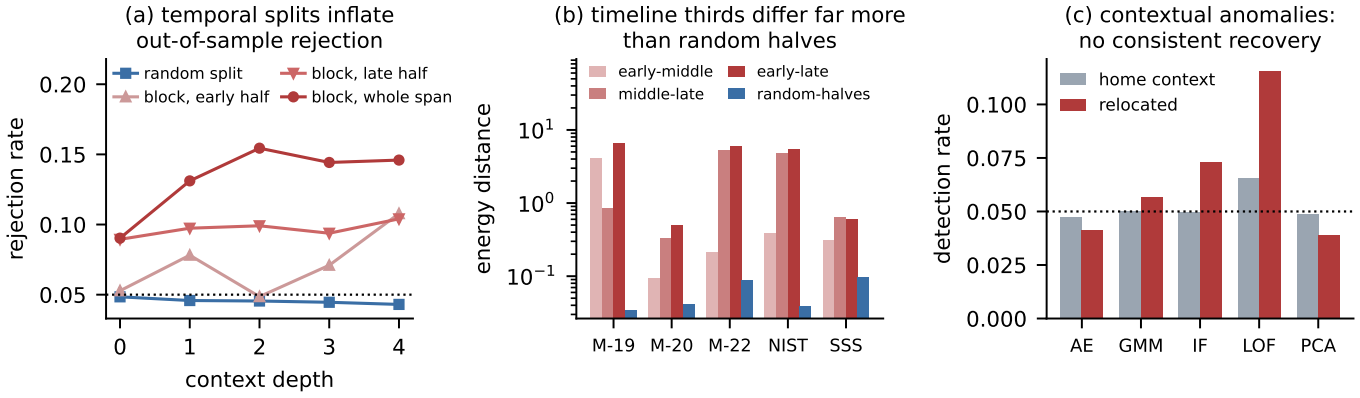



Figure 4: All panels use the NTFS supertimeline substrate. (a) Out-of-sample rejection rate on non-injected episodes when reference and test are split in different ways. The dotted line is the nominal $\alpha = 0.05$. (b) Energy distance between the behavioural feature distributions of pairs of timeline thirds, and between two random halves of the same exhibit, on a log scale. Exhibits are Magnet CTF 2019, 2020 and 2022 (M-19, M-20, M-22), the NIST Data Leakage case and Stolen Szechuan Sauce (SSS). (c) Rejection rate of non-injected episodes calibrated against their own context and against a different context of the same depth, by detector. The dotted line is the nominal $\alpha$.

*What an examiner can do.* None of the three tested reference strategies reliably repairs the problem within the exhibit. Examiners can nevertheless take two steps before building a review queue. First, the admissibility grid uses episode counts to establish whether a context supports the reporting level before any detector runs. If it does not, the options are a shallower context, a larger queue, or no levelled claim. The last is sometimes the only defensible option. Second, screening results should report incident multiplicity and its distribution across contexts. The loss depends on how much of the incident shares a context with the tested episode. Neither step improves detection, but both clarify what the result can support.

*Separating context from behaviour.* An earlier version of this study drew context axes from the behavioural vector and produced a large apparent absorption effect. Conditioning on an attribute that the detector also scores removes its signal by construction. A contextual forensic detector should identify the attributes used for contexts and scoring, and keep the two lists disjoint.

*The instrument.* Evaluation protocols for unsupervised outlier detection have been scrutinised before, mainly for measures and datasets [11]. We add a control for the composition of the reference population itself. The size-matched clean counterfactual does not repair the reference. Applied to any detector on any exhibit, it attributes a loss to contamination, conditioning, or reference size. Together with the transplant harness and admissibility grid, it provides a reusable evaluation protocol independent of our conclusions.

## 7. Limitations

*Exhibits are challenge images.* All five are public forensic challenge or scenario images, not casework. Their activity is planted and temporally compressed relative to a real intrusion. We therefore do not interpret absolute magnitudes as casework estimates. The mechanism can operate in either setting whenever incident siblings enter the calibration reference. This depends on reference construction rather than on how the activity was produced. The admissibility bound depends only on episode counts. Whether reference siblings reduce detection, and by how much, depends on their distribution across contexts and their scores relative to the tested episode and the rest of the reference. The prevalence and magnitude of the effect in casework remain unmeasured here.

*Constructed ground truth.* Injected anomalies are real episodes but not annotated incident activity. Exhibit identity is also partly predictable from the behavioural features. Every comparison is paired, with the injected set held fixed across arms. Section 5.4 reports checks on these concerns. Same-exhibit donors remove exhibit-identity shift. The answer-key-aligned analysis provides an ecological sanity check rather than an effect-size estimate.

*Episode granularity.* Results depend on the episode definition. Over inactivity gaps of 60/120/300 s the three larger exhibits are stable in admissibility, while the two smaller ones fall to zero coverage at some settings. The main size-matched contamination effect retains its direction and significance across these gaps. At $k = 50$, matched loss declines from 0.293 to 0.161 as episodes grow coarser. Fewer, larger episodes leave fewer contexts in which siblings can accumulate. This check includes only injections sharing a context with at least one sibling, since a lone injection has no sibling to mask it. Under that restriction, the main design gives 0.294, compared with 0.250 in Table 3, which averages over both cases.

*What the intervals cover.* Bootstrap intervals are clustered over (exhibit, donor, detector) with 3000 resamples. Only three exhibits are large enough to be recipients. The intervals therefore quantify variation within that design rather than across casework populations. They also condition on a single fit of each detector;

three of the five detectors are stochastic. Repeating the design of Section 5.2 with three fitting seeds gives size-matched losses of 0.238 to 0.266 at $k = 50$. Every interval excludes zero above $k = 1$.

*Scope.* We do not propose a detector and do not claim these obstacles are insurmountable. We claim they are present, practically consequential, and not accounted for in the forensic evaluations we study.

## 8. Conclusion

In a single post-incident exhibit, the examiner cannot identify a reference population known to be clean, representative and large enough. The three tested strategies fail in different ways. Context size bounds the attainable significance level. At levels motivated by forensic reporting, deep contexts become unusable. A size-matched counterfactual attributes the resulting loss to which episodes contaminate the reference rather than how many. The loss persists with no detectable change in magnitude when donor and recipient episodes come from the same exhibit. It also survives aggregation, reducing the probability of discovering any part of the incident by 0.085 to 0.132. Deeper contexts, searches across contexts and earlier timeline blocks do not reliably repair the loss. Earlier blocks also sacrifice exchangeability.

These findings do not preclude forensic anomaly detection. A forensic claim needs a defensible reference population before a better anomaly score can be useful. This prerequisite can be assessed before the first episode is reviewed.

## Data availability

The dataset is public [26]. The episode construction, injection harness, size-matched controls, prespecified analysis choices and all analysis scripts will be released on publication. These materials include a lightweight preflight utility for Plaso-derived timelines. For each significance level, it reports the deepest forensic context an exhibit can support. The submitted version omits the repository link to preserve anonymity.

## References


[1] Arp, D., Quiring, E., Pendlebury, F., Warnecke, A., Pierazzi, F., Wressnegger, C., Cavallaro, L., Rieck, K., 2022. Dos and don'ts of machine learning in computer security, in: 31st USENIX Security Symposium (USENIX Security 22), USENIX Association, Boston, MA. pp. 3971–3988.

[2] Barber, R.F., Candès, E.J., Ramdas, A., Tibshirani, R.J., 2023. Conformal prediction beyond exchangeability. The Annals of Statistics 51, 816 – 845. doi:10.1214/23-AOS2276.

[3] Bashari, M., Sesia, M., Romano, Y., 2025. Robust conformal outlier detection under contaminated reference data, in: Proceedings of the 42nd International Conference on Machine Learning, PMLR. pp. 3091–3141.

[4] Bates, S., Candès, E., Lei, L., Romano, Y., Sesia, M., 2023. Testing for outliers with conformal p-values. The Annals of Statistics 51, 149–178. doi:10.1214/22-AOS2244.

[5] Bergmann, P., Fauser, M., Sattlegger, D., Steger, C., 2019. Mvtec ad — a comprehensive real-world dataset for unsupervised anomaly detection, in: 2019 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), pp. 9584–9592. doi:10.1109/CVPR.2019.00982.

[6] Bhattacharyya, S., Ramdas, A., 2026. Change detection with conformal martingales: new optimal constructions, and suboptimality of existing methods. arXiv:2609.27179.

[7] Bollé, T., Casey, E., Jacquet, M., 2020. The role of evaluations in reaching decisions using automated systems supporting forensic analysis. Forensic Science International: Digital Investigation 34, 301016. doi:10.1016/j.fsidi.2020.301016.

[8] Breitinger, F., Studiawan, H., Hargreaves, C., 2025. SoK: Timeline based event reconstruction for digital forensics: Terminology, methodology, and current challenges. Forensic Science International: Digital Investigation 53, 301932. doi:10.1016/j.fsidi.2025.301932.

[9] Breunig, M.M., Kriegel, H.P., Ng, R.T., Sander, J., 2000. LOF: identifying density-based local outliers, in: Proceedings of the 2000 ACM SIGMOD International Conference on Management of Data, Association for Computing Machinery, New York, NY, USA. p. 93–104. doi:10.1145/342009.335388.

[10] Calikus, E., Nowaczyk, S., Bouguelia, M.R., Dikmen, O., 2022. Wisdom of the contexts: active ensemble learning for contextual anomaly detection. Data Mining and Knowledge Discovery 36, 2410–2458. doi:10.1007/s10618-022-00868-7.

[11] Campos, G.O., Zimek, A., Sander, J., Campello, R.J., Micenková, B., Schubert, E., Assent, I., Houle, M.E., 2016. On the evaluation of unsupervised outlier detection: measures, datasets, and an empirical study. Data Mining and Knowledge Discovery 30, 891–927. doi:10.1007/s10618-015-0444-8.

[12] Chandola, V., Banerjee, A., Kumar, V., 2009. Anomaly detection: A survey. ACM Computing Surveys 41, 1–58. doi:10.1145/1541880.1541882.

[13] Davies, L., Gather, U., 1993. The identification of multiple outliers. Journal of the American Statistical Association 88, 782–792. doi:10.1080/01621459.1993.10476339.

[14] Du, M., Li, F., Zheng, G., Srikumar, V., 2017. DeepLog: Anomaly detection and diagnosis from system logs through deep learning, in: Proceedings of the 2017 ACM SIGSAC Conference on Computer and Communications Security, Association for Computing Machinery, New York, NY, USA. p. 1285–1298. doi:10.1145/3133956.3134015.

[15] Du, X., Hargreaves, C., Sheppard, J., Anda, F., Sayakkara, A., Le-Khac, N.A., Scanlon, M., 2020. SoK: exploring the state of the art and the future potential of artificial intelligence in digital forensic investigation, in: Proceedings of the 15th International Conference on Availability, Reliability and Security, Association for Computing Machinery, New York, NY, USA. doi:10.1145/3407023.3407068.

[16] Emmott, A.F., Das, S., Dietterich, T., Fern, A., Wong, W.K., 2013. Systematic construction of anomaly detection benchmarks from real data, in: Proceedings of the ACM SIGKDD Workshop on Outlier Detection and Description, Association for Computing Machinery, New York, NY, USA. p. 16–21. doi:10.1145/2500853.2500858.

[17] Esposito, S., Peterson, G., 2013. Creating super timelines in windows investigations, in: IFIP International Conference on Digital Forensics, Springer. pp. 135–144. doi:10.1007/978-3-642-41148-9_9.

[18] Goh, J., Adepu, S., Junejo, K.N., Mathur, A., 2016. A dataset to support research in the design of secure water treatment systems, in: International conference on critical information infrastructures security, Springer. pp. 88–99. doi:10.1007/978-3-319-71368-7_8.

[19] Guðjónsson, K., 2010. Mastering the super timeline with log2timeline. SANS Institute .

[20] Hargreaves, C., Patterson, J., 2012. An automated timeline reconstruction approach for digital forensic investigations. Digital Investigation 9, S69–S79. doi:10.1016/j.diin.2012.05.006.

[21] Hawkins, D.M., 1980. Identification of outliers. volume 11. Springer.

[22] Khalid, Z., Iqbal, F., Saqib, M., 2025. Bridging knowledge gaps in digital forensics using unsupervised explainable ai. Forensic Science International: Digital Investigation 53, 301924. doi:10.1016/j.fsidi.2025.301924.

[23] Koizumi, Y., Kawaguchi, Y., Imoto, K., Nakamura, T., Nikaido, Y., Tanabe, R., Purohit, H., Suefusa, K., Endo, T., Yasuda, M., Harada, N., 2020. Description and discussion on dcase2020 challenge task2: Unsupervised anomalous sound detection for machine condition monitoring. arXiv:2006.05822.

[24] Kuang, Q., Xia, Y., 2026. Anytime-valid distribution shift detection via predictive rank martingales. arXiv:2609.00536.

[25] Liu, F.T., Ting, K.M., Zhou, Z.H., 2008. Isolation forest, in: 2008 Eighth IEEE International Conference on Data Mining, pp. 413–422. doi:10.1109/ICDM.2008.17.

[26] Marková, E., Sokol, P., Krišáková, S.P., Kováčová, K., 2024. Dataset of Windows operating system forensics artefacts. Data in Brief 55, 110693. doi:10.1016/j.dib.2024.110693.

[27] Mirsky, Y., Doitshman, T., Elovici, Y., Shabtai, A., 2018. Kitsune: An ensemble of autoencoders for online network intrusion detection, in: Network and Distributed Systems Security (NDSS) Symposium 2018. doi:10.14722/ndss.2018.23204.

[28] Ruff, L., Kauffmann, J.R., Vandermeulen, R.A., Montavon, G., Samek, W., Kloft, M., Dietterich, T.G., Müller, K.R., 2021. A unifying review of deep and shallow anomaly detection. Proceedings of the IEEE 109, 756–795. doi:10.1109/JPROC.2021.3052449.

[29] Song, X., Wu, M., Jermaine, C., Ranka, S., 2007. Conditional anomaly detection. IEEE Transactions on Knowledge and Data Engineering 19, 631–645. doi:10.1109/TKDE.2007.1009.

[30] Studiawan, H., Sohel, F., 2021. Anomaly detection in a forensic timeline with deep autoencoders. Journal of Information Security and Applications 63, 103002. doi:10.1016/j.jisa.2021.103002.

[31] Székely, G.J., Rizzo, M.L., 2013. Energy statistics: A class of statistics based on distances. Journal of Statistical Planning and Inference 143, 1249–1272. doi:10.1016/j.jspi.2013.03.018.

[32] Tibshirani, R.J., Foygel Barber, R., Candes, E., Ramdas, A., 2019. Conformal prediction under covariate shift, in: Advances in Neural Information Processing Systems, Curran Associates, Inc.

[33] Vovk, V., Gammerman, A., Shafer, G., 2005. Algorithmic learning in a random world. Springer.

[34] Vovk, V., Petej, I., Nouretdinov, I., Ahlberg, E., Carlsson, L., Gammerman, A., 2021. Retrain or not retrain: conformal test martingales for change-point detection, in: Proceedings of the Tenth Symposium on Conformal and Probabilistic Prediction and Applications, PMLR. pp. 191–210.

[35] Zhang, Y., Tan, Z., Luo, X., Liu, Y., 2026. Hierarchical reference sets for robust unsupervised detection of scattered and clustered outliers. IEEE Internet of Things Journal 13, 9350–9364. doi:10.1109/JIOT.2025.3644591.